Gioia Zheng

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Profile

Undergraduate at Sapienza University of Rome working on retrieval and evaluation systems for retrieval-augmented generation. Focus on full-pipeline construction (sparse + dense retrieval, reranking, generation), paired-bootstrap evaluation methodology, and reproducibility scaffolding.

Education

Sapienza University of Rome Sep 2022 – Present
B.Sc. in Applied Computer Science and Artificial Intelligence
Relevant coursework: TODO *4–6 items, e.g. Machine Learning, Algorithms, Information Retrieval, Statistics, Distributed Systems*

Experience

NLP Algorithm Engineer Intern Feb 2026 – May 2026
Microsoft — Natural Language Computing (Remote)

- **Token-F1 0.197 → 0.368** on MS MARCO dev/small (6,980 queries) by adding a cross-encoder reranker over dense top-100; 95% paired-bootstrap CIs on all four surface-form Δs strictly above 0.
- Built the end-to-end pipeline: BM25 and dense (SBERT/FAISS) retrieval, cross-encoder reranking, T5 generation, paired-bootstrap evaluation harness.
- +0.173 **DistilBERT-BERTScore Δ** on 3,000 paired examples ruled out a surface-form artefact; deterministic failure taxonomy attributed **∼90% of regressions to generator-side truncation**, not retrieval failure.
- **NLI-entailment Δ sign reversal (**−0.145**)** — the only metric in the pipeline whose paired Δ reverses sign — flagged as evidence for generator-capacity follow-up rather than decode-budget tuning.
- Reproducibility scaffolding: per-run manifest (git SHA, config hash, dependency-file hashes, environment fingerprint), pinned lockfile, 286-test pytest suite, CI on every push.

Projects

Handwritten OCR pipeline github.com/GioiaZheng/handwritten-ocr-system
PyTorch — CNN + BiLSTM + CTC

- Variable-length transcription pipeline on the IAM handwriting dataset: CNN feature extractor, BiLSTM encoder, CTC decoder.
- Train / dev / test split with held-out evaluation reporting CER and WER; numbers comparable across architecture sweeps.
- End-to-end training and inference scripts; thin web front-end for interactive evaluation.

LeetCode solutions and pattern notes github.com/GioiaZheng/Leetcode-Solutions
Python — GitHub Actions CI

- 98 solved problems, one per directory, with schema-validated metadata (id, difficulty, topics, status) regenerated into a topic index and a catalog on every push; CI fails if the generated docs drift out of sync.
- Pattern notes (sliding window, prefix sums, monotonic stack, dynamic programming, graphs, etc.) with cross-references to the corresponding problem directories.
- Quality pipeline: ruff lint, pytest on a curated subset, structural validation, multi-Python-version CI (3.11 / 3.12).

Chat service github.com/GioiaZheng/go-chat-system
Go (concurrent backend) + Vue + SQLite

- REST API documented under OpenAPI; Vue frontend embedded into the Go binary at build time for single-artifact deployment.
- Concurrent request handling with SQLite persistence; CI on every push (gofmt, go-vet, tests, OpenAPI lint, frontend build).

Open-source / Writing

TODO *one or two entries — e.g. a methodology blog post on surface-form vs. NLI-entailment disagreement in RAG evaluation; a HuggingFace dataset card; an issue or PR against pyserini / ir-measures / BEIR / sentence-transformers*

Skills

Primary	Python, PyTorch, BM25, FAISS, sentence-transformers, cross-encoder reranking, paired-bootstrap evaluation, NLI grounding, BERTScore
Working	Go (concurrent backends), Docker, Linux, GitHub Actions CI, OpenAPI, SQLite
Familiar	JavaScript / React Native, Vue
Languages	English (Professional), Italian (Bilingual), Chinese (Native), German (Beginner)